

OEAS 705/805 Advanced Environmental Data Science

Fall 2026

Week 1

Dr. Jessie Turner, jturners@odu.edu

Class times: OCNPS 320, T/Th, 11:00 - 12:15

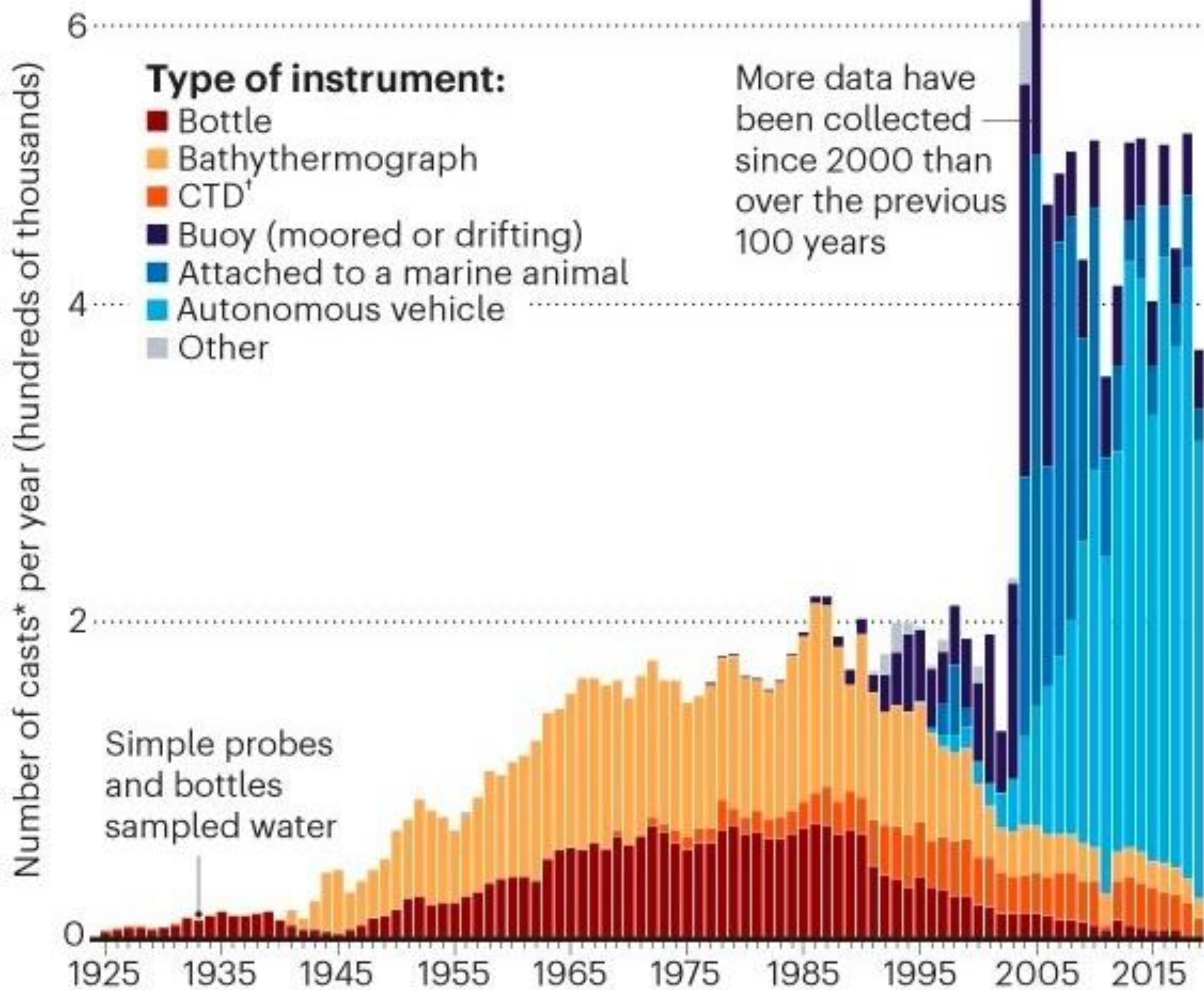
Office hours: After class T/Th,

And: OCNPS 411, M, 1:00 - 3:00



Intros

- Major and department
- Where from originally
- Other beings in your life (pets/kids/roommates)
- Research interests
- Coding/programming background
- Types of data you've worked with and/or generated
- Goals for this course
- Recommend a restaurant, coffee shop, or ice cream shop
- What you would be in another dimension



Learning objectives

1. Understand FAIR data principles (Findable, Accessible, Interoperable, and Reusable) and how to apply them.
2. Develop a working knowledge of existing environmental science databases and how to efficiently access data from them, including via APIs (Application Programming Interfaces).
3. Develop data analysis toolboxes using, but not limited to, python and shell scripts.
4. Understand and apply version control (e.g. git), environments (e.g. conda) and code repositories (e.g. GitHub) to manage and share code.
5. Understand the underlying principles of machine learning techniques for regression and classification, including supervised and unsupervised learning, and apply them to a targeted research question.
6. Understand the process of model evaluation and optimization and commonly used metrics for reporting model performance.

Final Project

- combine and apply the skills learned in class in the context of a real- world research problem.
- data analysis and visualization
- developing and evaluating machine learning models
- dataset(s) and general scope of their capstone project approved by the instructor prior to spring break.

Grading Summary

Homeworks	50%
Project report	20%
Project GitHub repository	15%
Project presentation	15%

Course Housekeeping

- Tools:**
- laptop
 - git
 - GitHub
 - conda
 - python
 - Interface (Virtual Studio Code or Jupyter Notebook).
 - Text editor (Sublime Text or BBEdit or other).

To Do:

- set up GitHub account before Thursday

check course info on GitHub:

https://github.com/jessiesturner/OEAS805_envdatasci

Questions?

What is Environmental Data Science?

Data Science

- An interdisciplinary field that uses statistics, programming, and domain knowledge to extract meaningful insights and predictions from data.

What is Environmental Data Science?

Data Science

- An interdisciplinary field that uses **statistics, programming, and domain knowledge** to extract meaningful insights and predictions from data.

“Domain knowledge”

- Your understanding of the science
- Can you tell whether your results are helpful vs. nonsense
- Context for what the data are showing

(Data) Science needs (good) data

- Open science framework
- FAIR data principles

“Open-source science requires a culture shift to a more inclusive, transparent, and collaborative scientific process, which will increase the pace and quality of scientific progress.”



OPEN (**TRANSPARENT**) SCIENCE

scientific process and results should be visible, accessible, and understandable

OPEN (**ACCESSIBLE**) SCIENCE

data, tools, software, documentation, and publications should be accessible to all (FAIR)



OPEN (**INCLUSIVE**) SCIENCE

process and participants should welcome participation by and collaboration with diverse people and organizations

OPEN (**REPRODUCIBLE**) SCIENCE

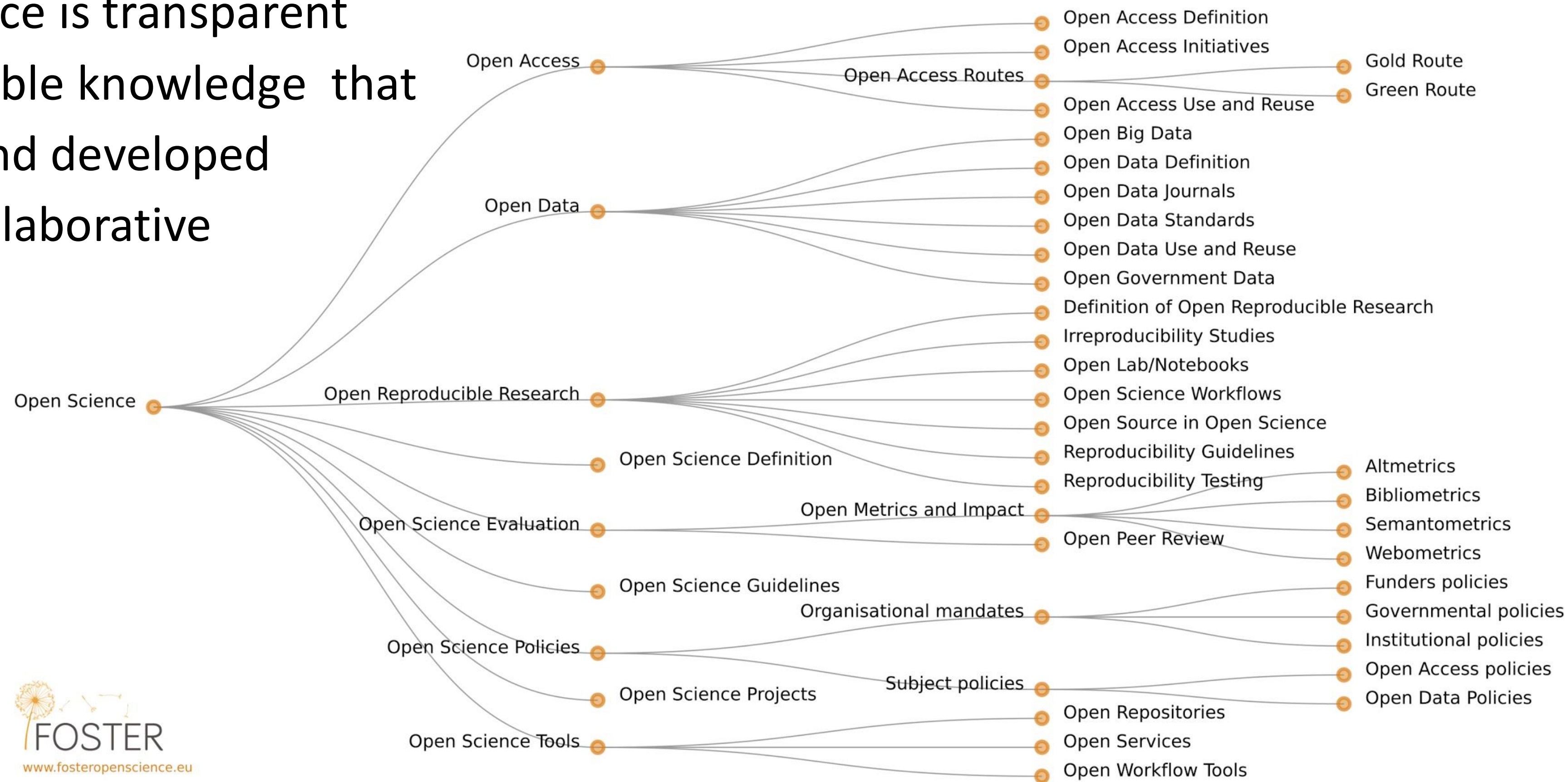
scientific process and results should be open such that they are reproducible by members of the community



Open Science Framework

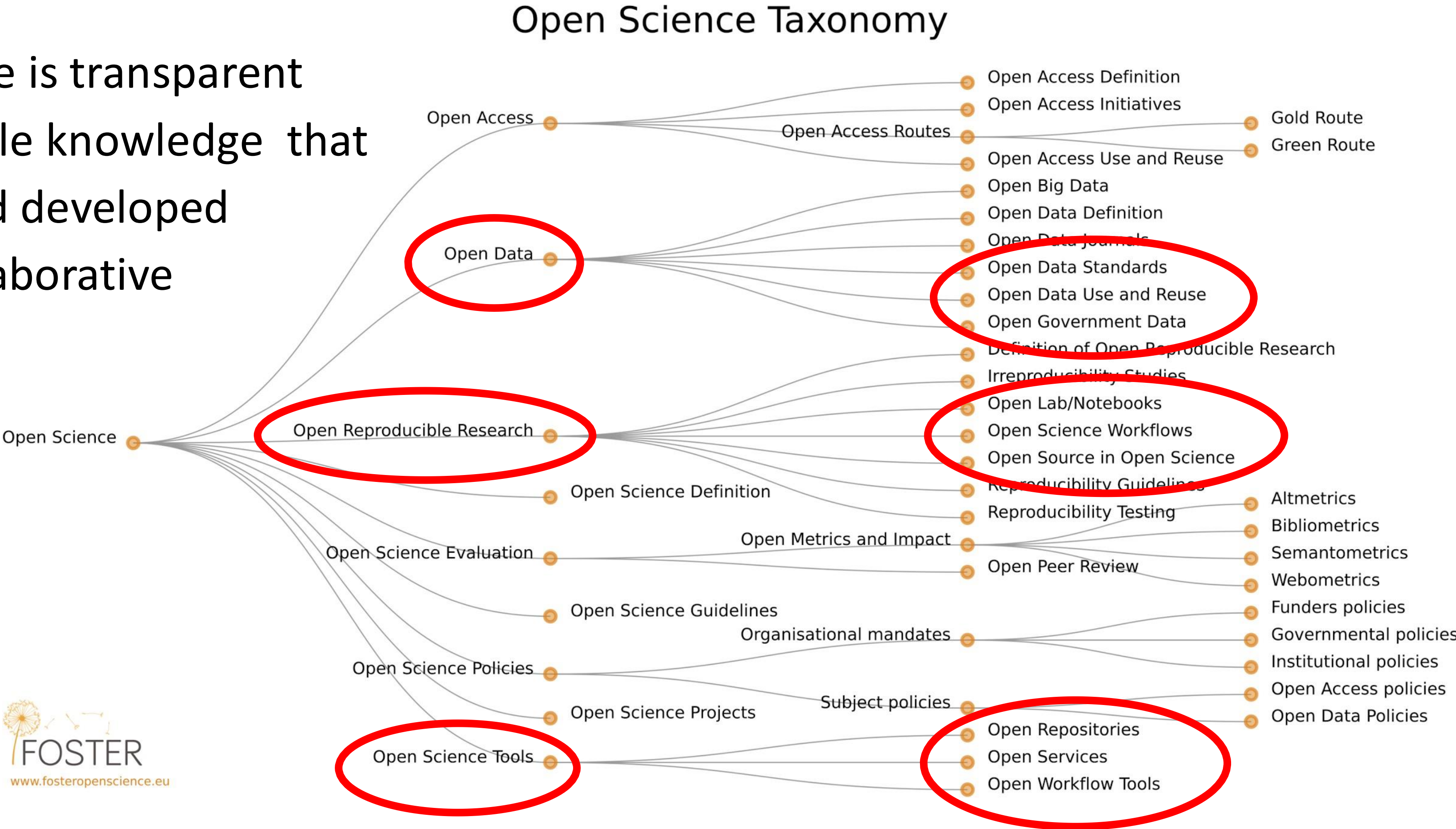
Open Science is transparent and accessible knowledge that is shared and developed through collaborative networks.

Open Science Taxonomy



Open Science Framework

Open Science is transparent and accessible knowledge that is shared and developed through collaborative networks.



FAIR data principles

A set of principles aimed at making it easier to find and re-use scholarly data.

Wilkinson, M.D., Dumontier, M., Aalbersberg, I.J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.W., da Silva Santos, L.B., Bourne, P.E. and Bouwman, J., 2016. The FAIR Guiding Principles for scientific data management and stewardship. *Scientific data*, 3, doi:[10.1038/sdata.2016.18](https://doi.org/10.1038/sdata.2016.18)

SCIENTIFIC DATA 

OPEN

SUBJECT CATEGORIES

» Research data

» Publication characteristics

Comment: The FAIR Guiding Principles for scientific data management and stewardship

Mark D. Wilkinson *et al.*[#]

There is an urgent need to improve the infrastructure supporting the reuse of scholarly data. A diverse set of stakeholders—representing academia, industry, funding agencies, and scholarly publishers—have come together to design and jointly endorse a concise and measurable set of principles that we refer to as the FAIR Data Principles. The intent is that these may act as a guideline for those wishing to enhance the reusability of their data holdings. Distinct from peer initiatives that focus on the human scholar, the FAIR Principles put specific emphasis on enhancing the ability of machines to automatically find and use the data, in addition to supporting its reuse by individuals. This Comment is the first formal publication of the FAIR Principles, and includes the rationale behind them, and some exemplar implementations in the community.

Received: 10 December 2015

Accepted: 12 February 2016

Published: 15 March 2016

FAIR data principles

A set of principles aimed at making it easier to find and re-use scholarly data.

What have we learned in the past 10 years?

Wilkinson, M.D., Dumontier, M., Aalbersberg, I.J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.W., da Silva Santos, L.B., Bourne, P.E. and Bouwman, J., 2016. The FAIR Guiding Principles for scientific data management and stewardship. *Scientific data*, 3, doi:[10.1038/sdata.2016.18](https://doi.org/10.1038/sdata.2016.18)

SCIENTIFIC DATA 

OPEN

SUBJECT CATEGORIES

» Research data
» Publication characteristics

Comment: The FAIR Guiding Principles for scientific data management and stewardship

Mark D. Wilkinson *et al.*[#]

There is an urgent need to improve the infrastructure supporting the reuse of scholarly data. A diverse set of stakeholders—representing academia, industry, funding agencies, and scholarly publishers—have come together to design and jointly endorse a concise and measureable set of principles that we refer to as the FAIR Data Principles. The intent is that these may act as a guideline for those wishing to enhance the reusability of their data holdings. Distinct from peer initiatives that focus on the human scholar, the FAIR Principles put specific emphasis on enhancing the ability of machines to automatically find and use the data, in addition to supporting its reuse by individuals. This Comment is the first formal publication of the FAIR Principles, and includes the rationale behind them, and some exemplar implementations in the community.

Received: 10 December 2015

Accepted: 12 February 2016

Published: 15 March 2016

FAIR data principles: definitions

The principles refer to three types of entities:

- **data** (or any digital object)
- **metadata**: information about that digital object **infrastructure**:
- a searchable resource that houses data and metadata

FAIR data principles

F = FINDABLE

A = ACCESSIBLE

I = INTEROPERABLE R

= REUSABLE

FAIR data principles

F = FINDABLE

The first step in (re)using data is to find them. Metadata and data should be easy to find for both humans and computers.

FAIR data principles

F = FINDABLE

A = ACCESSIBLE

Once the user finds the required data, she/he needs to know how can they be accessed, possibly including authentication and authorisation.

FAIR data principles

F = FINDABLE

A = ACCESSIBLE

I = INTEROPERABLE

The data usually need to be integrated with other data. In addition, the data need to interoperate with applications or workflows for analysis, storage, and processing.

FAIR data principles

F = FINDABLE

A = ACCESSIBLE

I = INTEROPERABLE R

= REUSABLE

The ultimate goal of FAIR is to optimise the reuse of data. To achieve this, metadata and data should be well- described so that they can be replicated and/or combined in different settings.

Open Science and FAIR data principles: discussion (15 mins)

Group 1: you have generated data

- What do you need to do to make it useful for other researchers?

Group 2: you are using someone else's data

- What do you need to know for it to be useable for your research?

Open Science/FAIR principles need to be baked into scientific workflows, can rarely be reverse engineered.

F = FINDABLE

A = ACCESSIBLE

I = INTEROPERABLE R

= REUSABLE

FAIR data principles

Check the “FAIRness” of a dataset:

<https://www.f-uji.net/>



F-UJI is a web service to programatically assess FAIRness of research data objects at the dataset level based on the FAIRsFAIR Data Object Assessment Metrics [↗](#)

[Click here to assess a dataset](#)

Before Thursday

- set up GitHub account
- check course info on GitHub:
https://github.com/jessiesturner/OEAS805_envdatasci
- read Wilkinson FAIR data paper (in readings folder on GitHub)

Thursday and every class from here on:

- bring laptop